

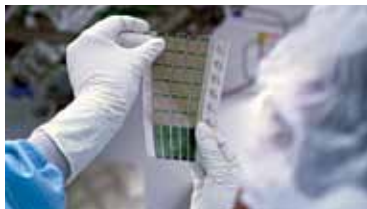


Next-gen responsive material learns from its past

Researchers at Aalto University have developed a new material that changes its electrical behaviour based on previous experience, effectively giving it a basic form of adaptive memory. Responsive materials have become common in a range of applications, from glasses that darken in sunlight to drug delivery systems. However, existing materials always react in the same way each time. Researchers synthesized micrometer-sized magnetic beads which were then stimulated by a magnetic field. When the magnet was on, the beads stacked up to form pillars. The strength of the magnetic field affects the shape of the pillars, which in turn affects how well they conduct electricity. The system's memory allows it to behave in a way that resembles rudimentary learning. When they exposed the beads to a quickly pulsing magnetic field, the material became better at conducting electricity, whereas slower pulsing made it conduct poorly.

Paper-thin solar cells turn any surface into a power source

Researchers at MIT have designed an ultralight fabric solar cell that can quickly and easily turn any surface into a power source. The cells are one-hundredth the weight of



MIT researchers have developed a scalable fabrication technique to produce ultrathin, lightweight solar cells that can be stuck onto any surface (Credit: Melanie Gonick, MIT)

conventional solar panels but generate 18 times more power per kilogram. As they are thin and lightweight, they can be laminated onto many different surfaces. Researchers used nanomaterials in the form of printable electronic inks to

produce these solar cells. Then they were coated using a slot-die coater, which deposits layers of electronic materials onto a releasable substrate that is only 3-micron thick. Using screen printing, an electrode was deposited on the structure to complete the solar module. The 15-micron thick printed module can be peeled off the plastic substrate, forming an ultralight solar device. When the device was tested, it could generate 730 watts of power per kilogram while freestanding and about 370 watts per kilogram if deployed on the high-strength Dyneema fabric.

Building materials atom by atom being made possible

Researchers have used deep reinforcement learning to steer atoms into a lattice shape with a view to building new ma-

terials or nanodevices. Such precise movement of atoms required to design very small devices based on single atoms is important for nanodevices like transistors or memories. "Give it a goal and it will do it. It can solve problems that humans don't know how to solve." Building new materials atom-by-atom, rather than through traditional chemical techniques, may also reveal interesting properties related to superconductivity or quantum states.

Telepathic device enables tactile sensation from distance

Researchers have developed a telepathic device that can be attached to the fingertip like a sticker and leverages the unique vibration pattern of touch and texture. By having this close contact with the skin, it is able to provide a more vivid tactile experience and bridge the immersion hindrance of other similar devices. Researchers used a piezoelectric element and an ultra-thin flexible substrate, which was able to precisely integrate microscopic sensors and actuators of less than 1mm on the substrate. The high-resolution composite sensor, which is finely structured at 1.8mm intervals, can feel tactile patterns in a wide frequency range of 1 ~ 1,000Hz. The device when tested was able to simultaneously measure both slow-changing pressure (static pressure) and fast-changing pressure (dynamic pressure).

Soft robot heals like human skin

Researchers at Cornell University have designed a soft robot capable of healing itself using light for dynamic sensing. They have developed stretchable fibre-optic sensors for creating soft robots and its components—from skin to wearables. The first



Researchers installed SHeaLDS in a soft robot resembling a four-legged starfish and equipped with feedback control. After the researchers punctured one of its legs, the robot was able to detect the damage and self-heal the cuts (Credit: Cornell University)

step for self-healing to work is that the robot must be able to identify something that needs to be fixed. Researchers pioneered a technique using fibre-optic sensors coupled with LED lights capable of detecting minute changes on the surface of the robot. When combined with a polyurethane urea elastomer, these sensors showcase rapid healing and strength. The resulting SHeaLDS (self-healing light guides for dynamic sensing) provides a damage-resistant soft robot that can self-heal from cuts at room temperature without any external intervention. Researchers developed and punctured one of its legs six times, after which the robot was every time able to detect the damage and self-heal each cut in about a minute.